Practical Work 2

Software Development Life Cycle and Requirements

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Task 1 The six phases

No.	Phase	Correct output	What goes wrong if skipped
1	Requirement gathering	SRS document signed by the customer	The team may build functions that users did not ask for and miss essential rules.
2	Feasibility study	A report stating whether the project is approved	The university may start a project that is too expensive, too slow, technically impossible or unlawful.
3	Design	HLD and LLD documents	Developers may create incompatible modules, weak data structures and an interface that does not support the workflow.
4	Coding	Source code and unit tests	There is no executable system to test or deploy.
5	Testing	Test report and list of defects	Payment, ordering and load defects may reach students and cafeteria staff.
6	Deployment and maintenance	Working system and a user manual	Users cannot use the product reliably, and defects or changing needs remain unresolved.

Task 2 The cost of a late mistake

2.1 Changes required during testing

• Correct the threshold in the SRS from 5,000,000 soum to 500,000 soum and review related rules.

• Change the delivery-fee calculation in the source code and unit tests.

• Update test cases, expected results, user-interface messages and any help text that shows the wrong threshold; then repeat regression testing.

2.2 Damage money cannot repair

Customer trust and the shop's reputation can be damaged. Customers who were charged incorrectly may stop using the shop even after refunds are issued.

2.3 Cheapest phase

The error should be caught during requirement gathering and requirements review at a relative cost of 1. The analyst would read the numeric rule back to the customer and compare the written SRS with the interview note before approval.

Task 3 Functional or non-functional

No.	Type	Quality if N	Reason
1	F	-	Shows today's menu.
2	N	Performance	Sets response time and concurrent load.
3	F	-	Provides card payment.
4	N	Security	Protects card data.
5	F	-	Sends a ready notification.
6	N	Usability	Specifies supported interface languages.
7	F	-	Creates a unique daily order number.
8	N	Portability	Specifies browsers and devices.
9	F	-	Shows unprepared orders.
10	N	Reliability	Limits downtime.

No.	Type	Quality if N	Reason
11	F	-	Provides a sales count by dish.
12	N	Maintainability	Limits the effect of a future change.

Task 4 Rewritten requirements

ID	Corrected requirement	Test
NFR-01	The menu page shall load within 2 seconds for at least 1,000 simultaneous students on the university Wi-Fi.	Open the page with a 1,000-user load test and verify every response is at most 2 seconds.
NFR-02	A first-year student shall find and add a dish to an order in no more than 3 clicks after opening the menu.	Ask 5 first-year students to add a named dish and count clicks; each must use at most 3.
NFR-03	The system shall support at least 2,000 registered students and 1,000 simultaneous active sessions.	Create 2,000 accounts, run 1,000 sessions and verify successful menu and order operations.
NFR-04	The system shall transmit payment data only through TLS 1.3 and shall store no card number or CVV in the cafeteria database.	Inspect network traffic during a payment and search all database fields for card number and CVV.
FR-11	The system shall show a paid order on the kitchen screen within 5 seconds after payment confirmation.	Pay for 10 orders and verify each appears within 5 seconds.
FR-12	The system shall produce by 18:05 a daily report containing total orders, portions sold for every dish and total revenue in soum.	Place 10 known orders, generate the report after 18:00 and compare all totals with the orders.

Task 5 Requirements analysis

5.1 Requirements list

ID	Requirement
FR-01	The system shall allow a student to sign in with one university account.
FR-02	The system shall show the menu for the current day with the price of every dish.
FR-03	The system shall allow a student to add up to 5 dishes to one order.
FR-04	The system shall allow payment by bank card or payment at the counter.
FR-05	The system shall assign every order a number unique within that calendar day.
FR-06	The system shall show every confirmed order on the kitchen screen within 5 seconds.
FR-07	The system shall allow kitchen staff to mark an order as ready.
FR-08	The system shall notify the student within 10 seconds after the order is marked ready.
FR-09	The system shall allow cancellation only while an order has the status Not started.
FR-10	The system shall produce a daily report of portions sold for every dish and total revenue.
NFR-01 Performance	The menu page shall load within 2 seconds for 1,000 simultaneous students on university Wi-Fi.
NFR-02 Security	Payment data shall use TLS 1.3 and card numbers and CVVs shall not be stored by the cafeteria system.
NFR-03 Usability	The interface shall be available in Uzbek, Russian and English.
NFR-04 Reliability	Downtime shall not exceed 10 minutes during one working day from 08:00 to 18:00.
NFR-05 Maintainability	An authorized manager shall add a dish category through the admin screen without changing source code.
NFR-06 Portability	Student ordering functions shall work in the two latest major versions of Chrome and Safari and on university tablets.

5.2 Collection techniques

Technique	Source	What will be learned
Interview	Cafeteria manager, cashier and cook	Payment rules, kitchen workflow, reporting needs and exceptions.
Questionnaire	300-500 students	Expected usage, preferred payment method, languages and notification preferences.

Technique	Source	What will be learned
Observation	Cafeteria during two busy breaks	Actual queue time, bottlenecks, collection process and peak order volume.
Document analysis	Menus, price lists, receipts and daily cash reports	Dish categories, prices, taxes, sales fields and reconciliation rules.
Prototyping	10 students and 3 staff members	Whether ordering, kitchen and manager screens are understandable before coding.

First choice: observation. It reveals the real peak-time workflow and bottlenecks without relying on memory, so later interviews can ask precise questions about events already seen.

5.3 Questions not answered

• What happens to a paid order that the student never collects?

• When is a refund allowed and how long may it take?

• How are allergies and ingredients shown and confirmed?

• Who may change prices, dishes and the number of available portions?

• What should happen when the internet or payment provider is unavailable?

5.4 Self-check

ID	Clear	Testable	Complete	Consistent	Necessary	Traceable
FR-01	yes	yes	yes	yes	yes	yes
FR-06	yes	yes	yes	yes	yes	yes
NFR-01	yes	yes	yes	yes	yes	yes

No correction is required: the selected requirements identify the actor or system behaviour, contain measurable conditions where needed, do not conflict with the case, support a stated stakeholder need and can be traced to authentication, kitchen processing and peak-load performance.

Task 6 Bonus

6.1 FR-20 contradicts FR-21. If cooking starts immediately after payment, cancellation at any moment before collection can waste prepared food and cannot always reverse the kitchen action.

6.2 NFR-09 is not a software requirement. It is a business goal for cafeteria operations and should be measured in the project plan, not imposed as an absolute software behaviour.

6.3 Corrected pair: FR-20 The student shall be able to cancel an order only while its status is Not started. FR-21 The kitchen shall change a confirmed order to Preparing before cooking, after which cancellation shall be disabled.